

Recap of Week I

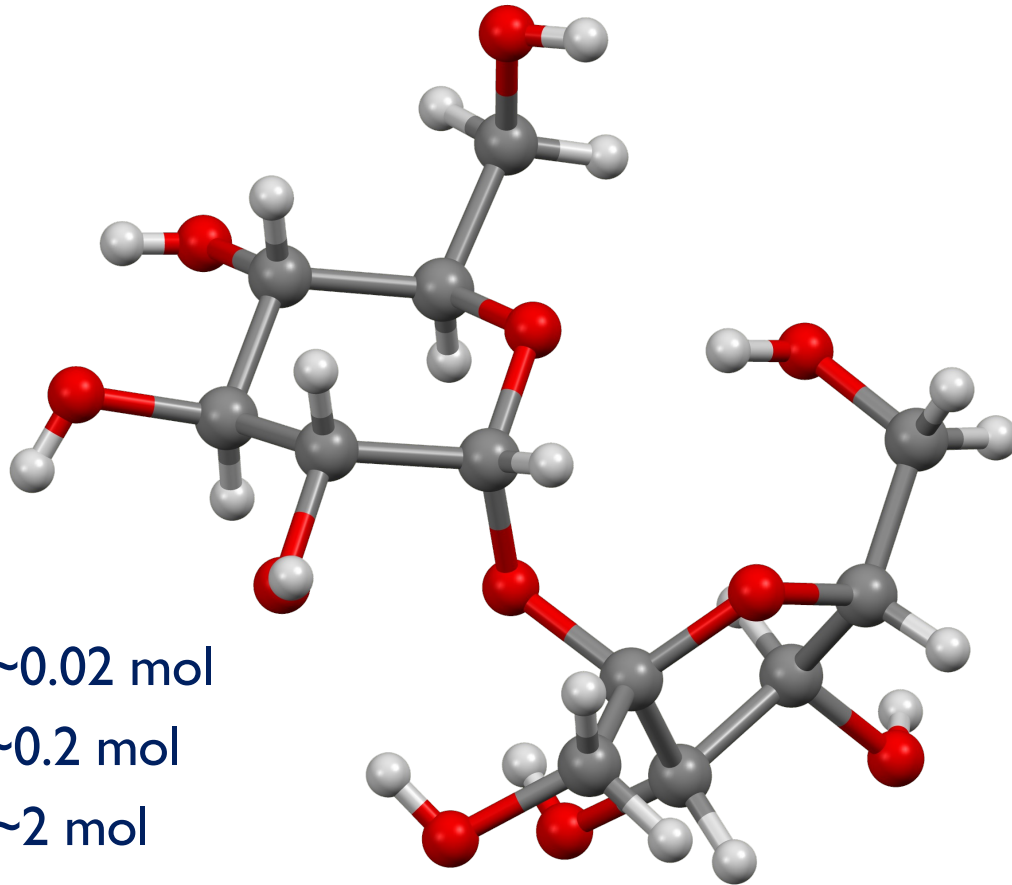
26021 Chemistry (Polytechnical Foundation)

Professor Kasper S. Pedersen

2026

What is 1 mol?

How many mol are there approximately in a 1 kg Goldbears? (77% sugar)



- A: ~0.02 mol
- B: ~0.2 mol
- C: ~2 mol
- D: ~2 mol
- E: ~20 mol

Hydrogen	Carbon	Nitrogen	Oxygen
1 H	6 C	7 N	8 O
1.0080	12.011	14.007	15.999



Let's enjoy a snack



- Balance the equation
- How much KClO_3 do I need to combust 1 kg of Goldbears?

Solution

$$m(\text{C}_{12}\text{H}_{22}\text{O}_{11}) = 770 \text{ g}$$

$$M(\text{C}_{12}\text{H}_{22}\text{O}_{11}) = 342 \text{ g mol}^{-1}$$

$$n(\text{C}_{12}\text{H}_{22}\text{O}_{11}) = m/M = 2.25 \text{ mol}$$



The combustion follows:

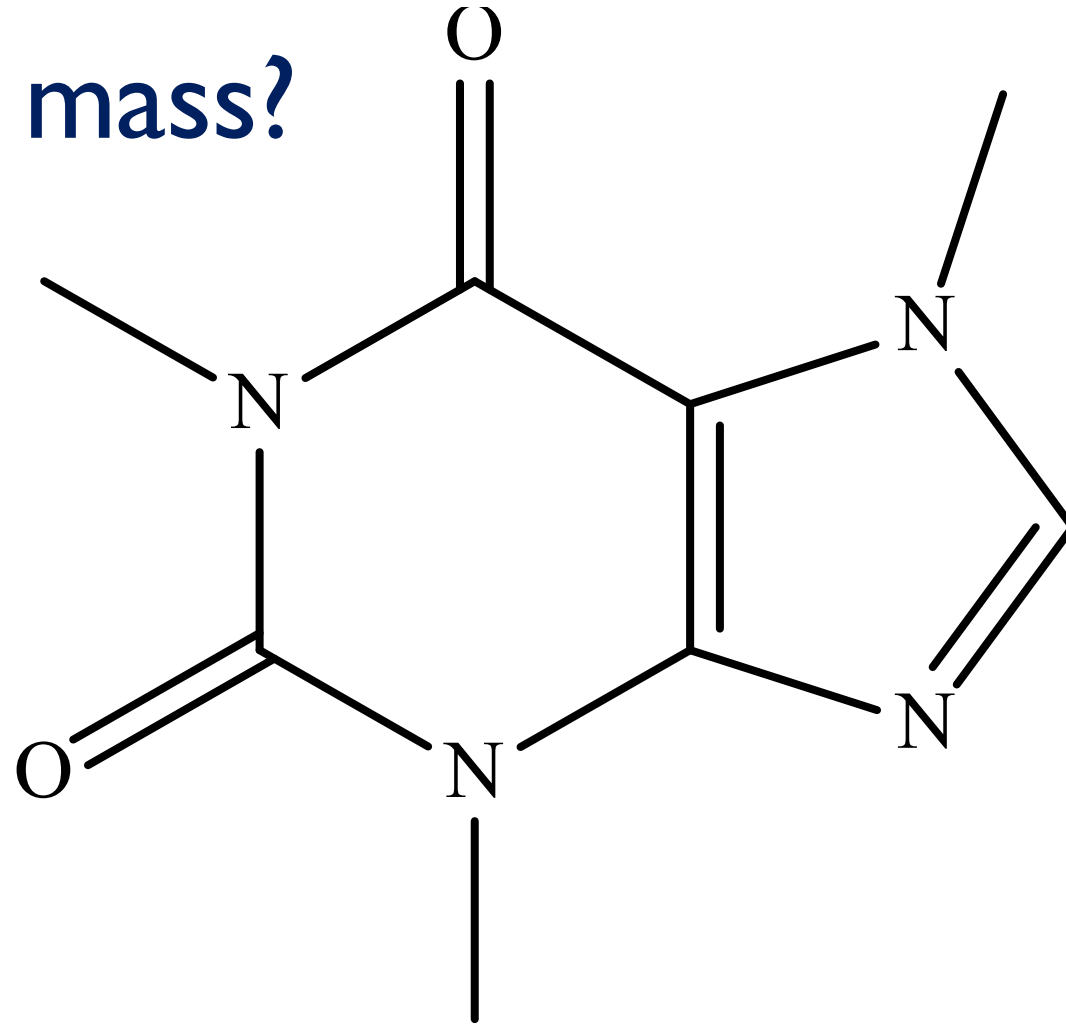


Thus, for a complete reaction we need

$$n(\text{O}_2) = 12 \times n(\text{C}_{12}\text{H}_{22}\text{O}_{11}) = 27.0 \text{ mmol}$$

This requires $n(\text{KClO}_3) = n(\text{O}_2)/(3/2) = \underline{18.0 \text{ mol (2.21 kg)}}$

Test:
Molar mass?



caffeine

Lecture 2: Reactions in aqueous solutions

26021 Chemistry (Polytechnical Foundation)

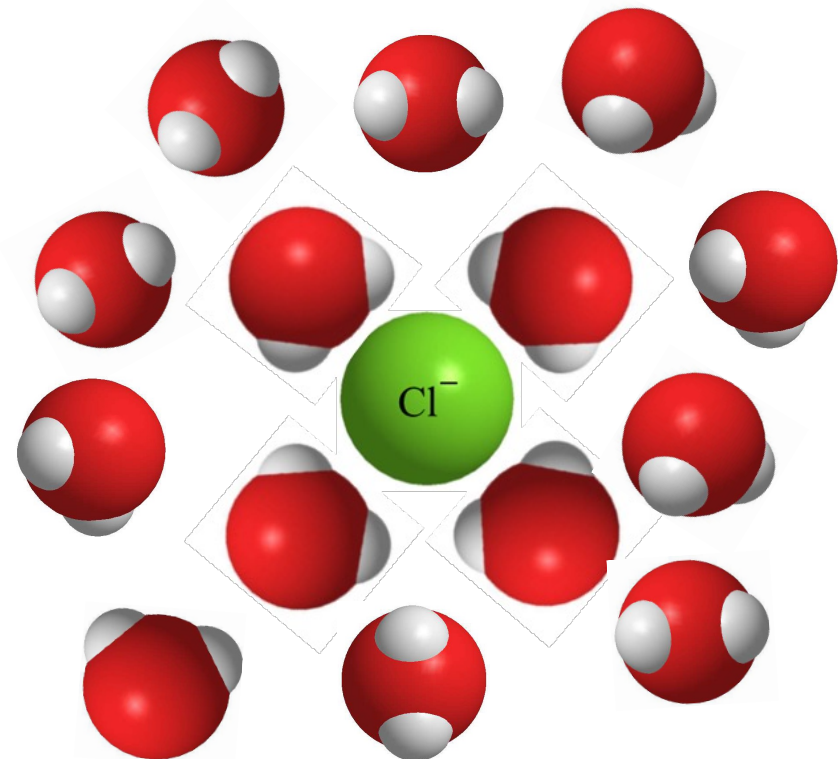
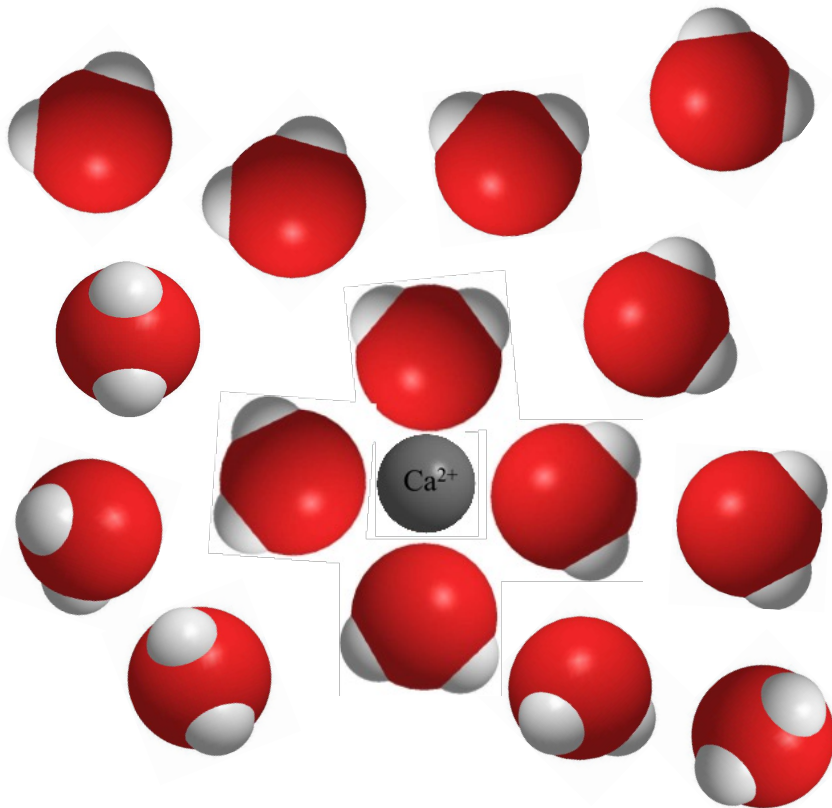
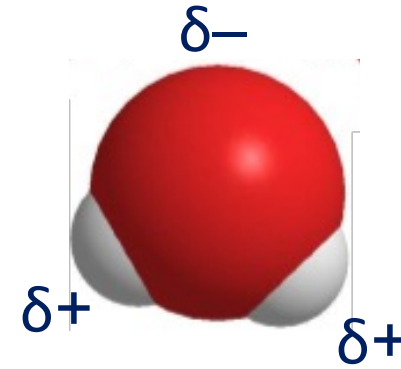
Professor Kasper S. Pedersen

2026

Solvation (hydration in case of H_2O)

Positive ions interact with **negative** parts of the water molecule.

Negative ions interact with **positive** parts of the water molecule.

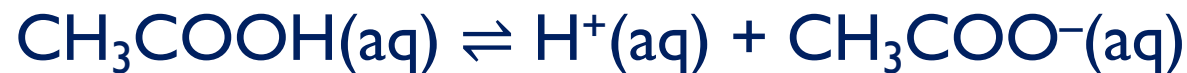


Properties of *electrolytes*

Strong electrolytes are completely ionized:



Weak electrolytes are only partly ionized:



Thus, an *equilibrium* (indicated by $\rightleftharpoons$) arises with all three species present.

The precise amount is calculated from the acid strength constant (K_a , more in a later lecture).

Concentration measures

Molarity given as number of mol per volume,
typically as $\text{mol L}^{-1} = \text{M}$.

Our fundamental equations now can be summarized as:

$$n = m/M$$

$$c = n/V$$

Exercise: What is the concentration of water in water? 🤔🤪

Concentration measures

Whilst c designates the *formal* concentration, we use $[]$ to indicate the *actual* concentration.

Exercise: What are the actual concentrations of Ca^{2+} and Cl^- in a 0.1 M solution of calcium chloride?

Electrolytes - examples

strong electrolyte	weak electrolyte	nonelectrolyte
HCl	CH ₃ COOH	(NH ₂) ₂ CO (urea)
HNO ₃	HF	CH ₃ OH (methanol)
HClO ₄	HNO ₂	C ₂ H ₅ OH (ethanol)
H ₂ SO ₄ *	NH ₃	C ₆ H ₁₂ O ₆ (glucose)
NaOH	H ₂ O [⊗]	C ₁₂ H ₂₂ O ₁₁ (sucrose)
Ba(OH) ₂		
Ionic compound		
* H ₂ SO ₄ : two ionizable H ⁺ ions		
⊗ Pure water: an extremely weak electrolyte.		

In pure water $[H^+] = [OH^-] = 10^{-7} \text{ M}$

Precipitation reactions

The solubility of a salt is determined by the competition between lattice energy and solvation energy.

For instance, the salts $\text{Pb}(\text{NO}_3)_2$ and KI are both highly soluble salts in water. What happens upon mixing?

Electrolytes - examples

Solubility Rules

associated with solubility

Li^+ , Na^+ , K^+ , NH_4^+ ,

NO_3^- , $\text{C}_2\text{H}_3\text{O}_2^-$

Cl^- , Br^- , I^-

SO_4^{2-}

exceptions

none

none

if paired with Ag^+ , Hg_2^{2+} , $\text{Pb}^{2+} \Rightarrow$ insoluble

if paired with Sr^{2+} , Ba^{2+} , Pb^{2+} , $\text{Ca}^{2+} \Rightarrow$ insoluble

associated with insolubility

OH^- , S^{2-}

exception

if paired with Li^+ , Na^+ , K^+ , $\text{NH}_4^+ \Rightarrow$ soluble

if S^{2-} paired with Ca^{2+} , Sr^{2+} , $\text{Ba}^{2+} \Rightarrow$ soluble

if OH^- paired with Ca^{2+} , Sr^{2+} , $\text{Ba}^{2+} \Rightarrow$ slightly soluble

CO_3^{2-} , PO_4^{3-}

if paired with Li^+ , Na^+ , K^+ , $\text{NH}_4^+ \Rightarrow$ soluble

Redox reactions

Redox = reduction-oxidation

Redox reaction: a chemical reaction where electrons are transferred between reactants

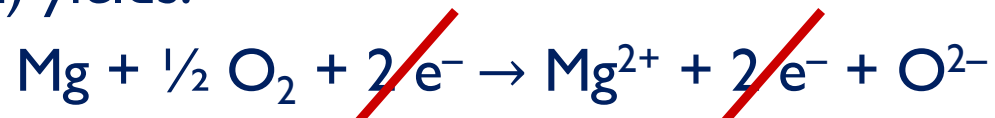
- Example, burning magnesium:



Which can be decomposed into two *half-cell* reactions:



(1) + (2) yields:



Activity (electrochemical) series

”preciousness” increases

Li $\rightarrow$ Li ⁺ + e ⁻	H ₂ evolves on contact with cold water
K $\rightarrow$ K ⁺ + e ⁻	
Ba $\rightarrow$ Ba ²⁺ + 2e ⁻	
Ca $\rightarrow$ Ca ²⁺ + 2e ⁻	
Na $\rightarrow$ Na ⁺ + e ⁻	
Mg $\rightarrow$ Mg ²⁺ + 2e ⁻	H ₂ evolves on contact with steam
Al $\rightarrow$ Al ³⁺ + 3e ⁻	
Zn $\rightarrow$ Zn ²⁺ + 2e ⁻	
Cr $\rightarrow$ Cr ³⁺ + 3e ⁻	
Fe $\rightarrow$ Fe ²⁺ + 2e ⁻	
Cd $\rightarrow$ Cd ²⁺ + 2e ⁻	H ₂ evolves on contact with acids
Co $\rightarrow$ Co ²⁺ + 2e ⁻	
Ni $\rightarrow$ Ni ²⁺ + 2e ⁻	
Sn $\rightarrow$ Sn ²⁺ + 2e ⁻	
Pb $\rightarrow$ Pb ²⁺ + 2e ⁻	no H ₂ generated on contact with water and acids
H ₂ $\rightarrow$ 2H ⁺ + 2e ⁻	
Cu $\rightarrow$ Cu ²⁺ + 2e ⁻	
Ag $\rightarrow$ Ag ⁺ + e ⁻	
Hg $\rightarrow$ Hg ²⁺ + 2e ⁻	
Pt $\rightarrow$ Pt ²⁺ + 2e ⁻	
Au $\rightarrow$ Au ³⁺ + 3e ⁻	



Oxidation state

Oxidation states (OS) are formal book-keeping – but they are useful for, e.g., balancing reaction schemes.

Rules for OS:

- 1) OS = 0 for free elements.
- 2) In monoatomic ions, OS = charge.
- 3) OS = $-II$ for oxygen in compounds (except O_2^{2-} and O_2^-)
- 4) OS = $+I$ for hydrogen, except when bound to metals
- 5) Halogens have OS = $-I$, except when bound to oxygen.
- 6) The sum of the OS in a molecule/ion must equal the charge of the molecule/ion.

Oxidation state - examples



Redox reactions – how to balance?

Conservation of electrons, atoms, and charge

Procedure:

- (1) Identify half-cell reactions
- (2) Balance the electron count by a common denominator and add half-reactions
- (3) Balance charges using H^+ in acidic solution, OH^- in basic solution (or either in neutral solution). Finally, add H_2O to balance.

Redox reactions - example

Sulfite (SO_3^{2-}) reacts with permanganate (MnO_4^-) in basic solution:



Let's balance! 🥰💃🕺

Redox reactions - examples

(I) Identify half-cell reactions



Half-cell reactions:



Redox reactions - examples

(2) Balance the electron count

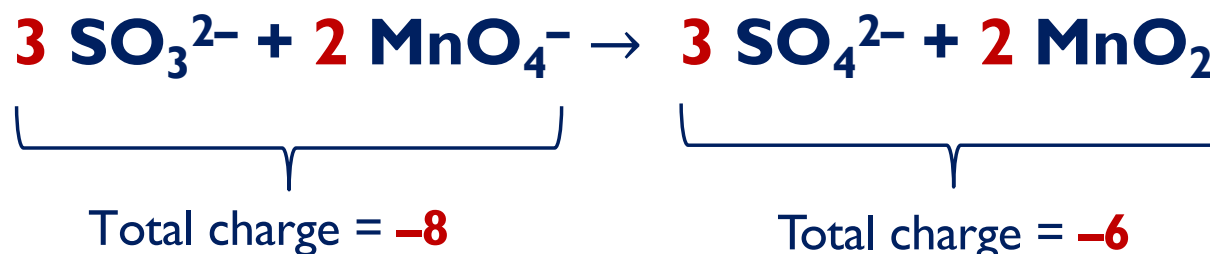


Half-cell reactions:

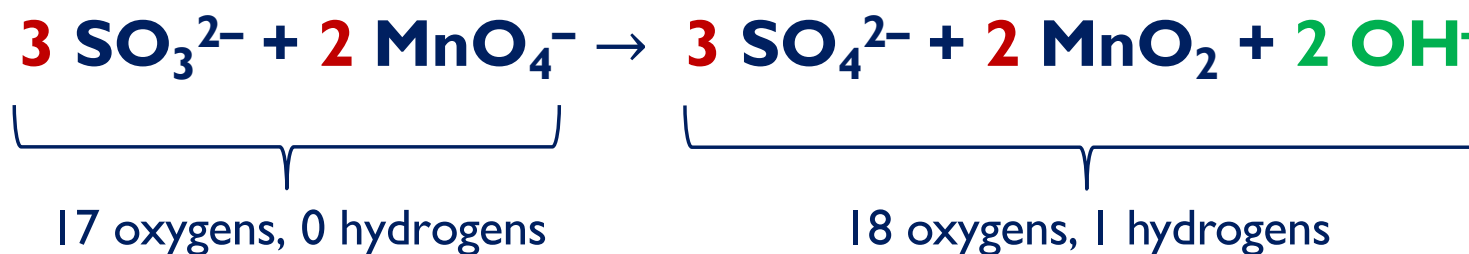


Redox reactions - examples

(3) Balance the charges



Difference of 2 in charge is compensated by adding 2 OH⁻:



Difference of oxygen and hydrogen is compensated by adding 1 H₂O:



We do this short – subject is
treated in more detail in
Physics (Polytechnic
Foundation)

Lecture 3: Gases

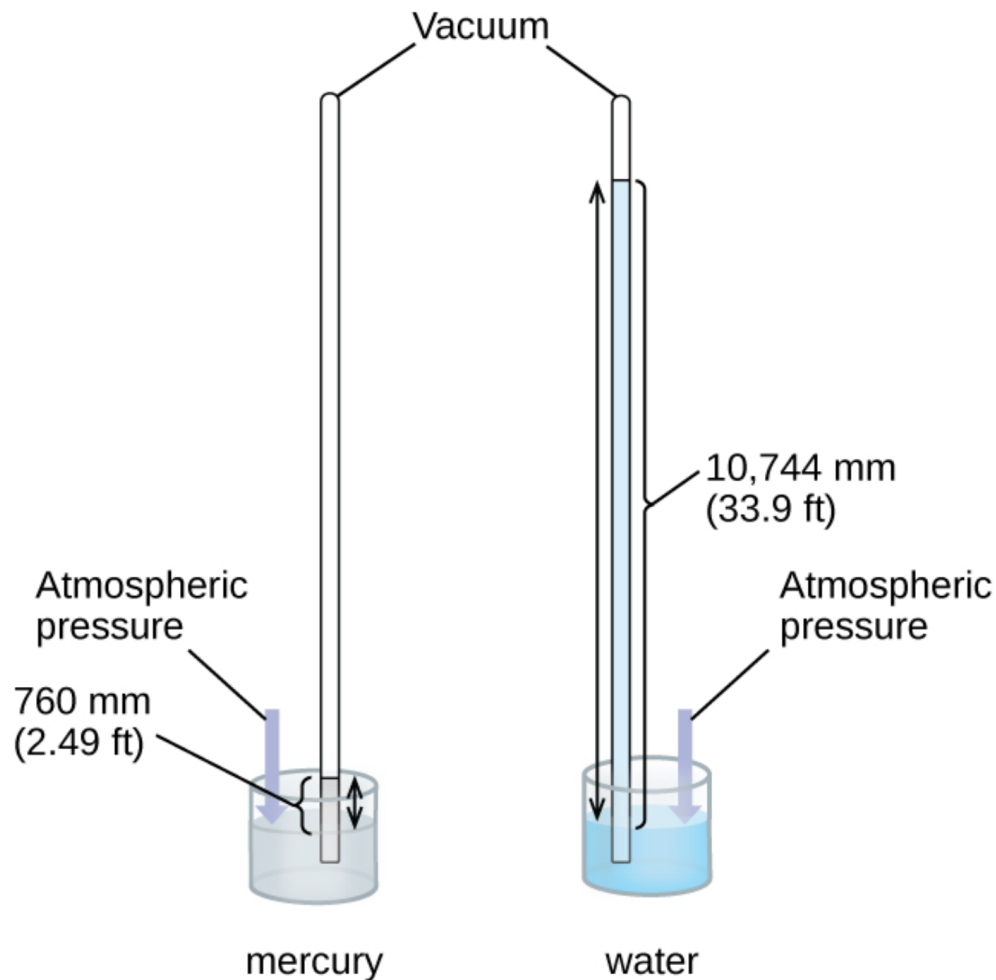
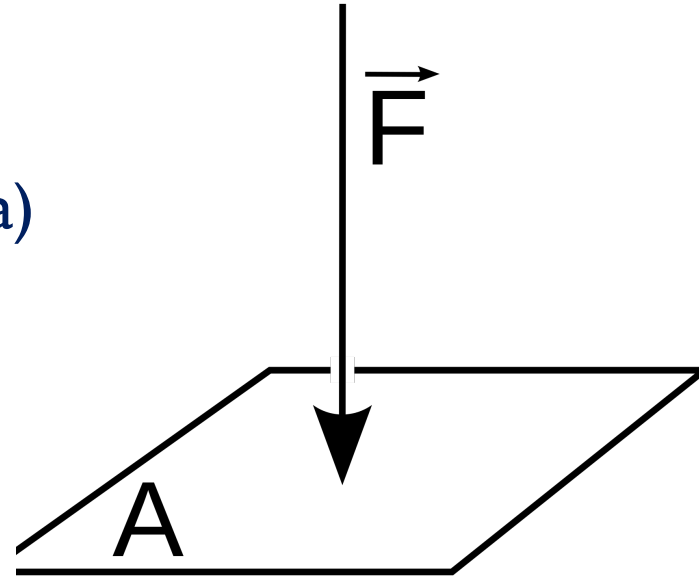
26021 Chemistry (Polytechnical Foundation)

Professor Kasper S. Pedersen

2026

Pressure, p

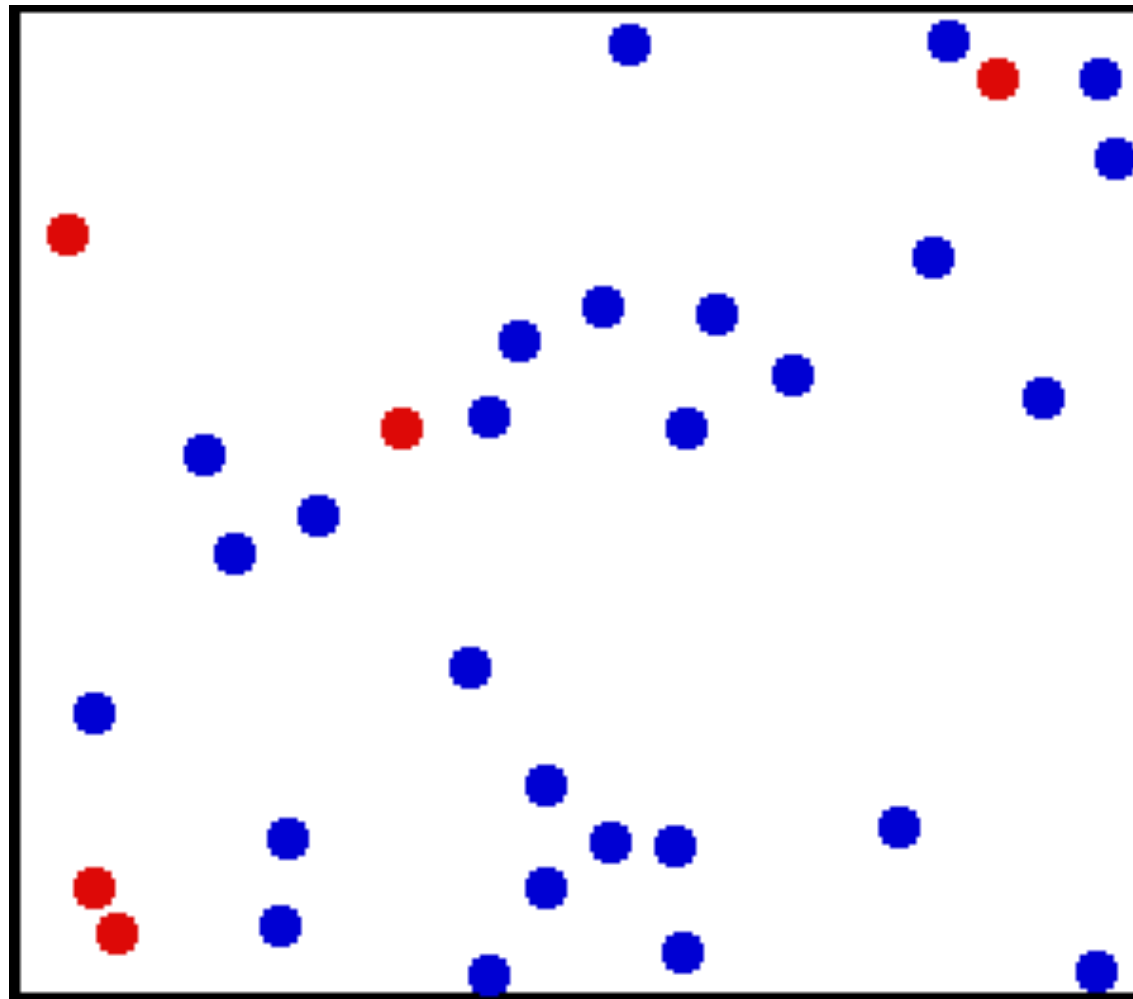
- Pressure: $p = F/A$ (force per area)
- Units: $1 \text{ N/m}^2 = 1 \text{ Pa}$



On earth: $1 \text{ atm} = 101325 \text{ Pa}$
 $1 \text{ bar} = 10^5 \text{ Pa} = 10^3 \text{ hPa}$
 $760 \text{ mmHg} = 1 \text{ atm}$

Number and velocity determine p

- More gas molecules $\rightarrow$ proportionally higher pressure
- Higher velocity gas molecules $\rightarrow$ proportionally higher pressure



Gas laws

At constant T :	$p \propto V^{-1}$	Boyle's law (1662)
At constant p :	$V \propto T$	Charles' law (1780)
At constant V :	$p \propto T$	Gay-Lussac's law (1780)

Ideal gas law: *For most gasses (at temperatures far from condensation and at low pressures):*

$$pV = nRT$$

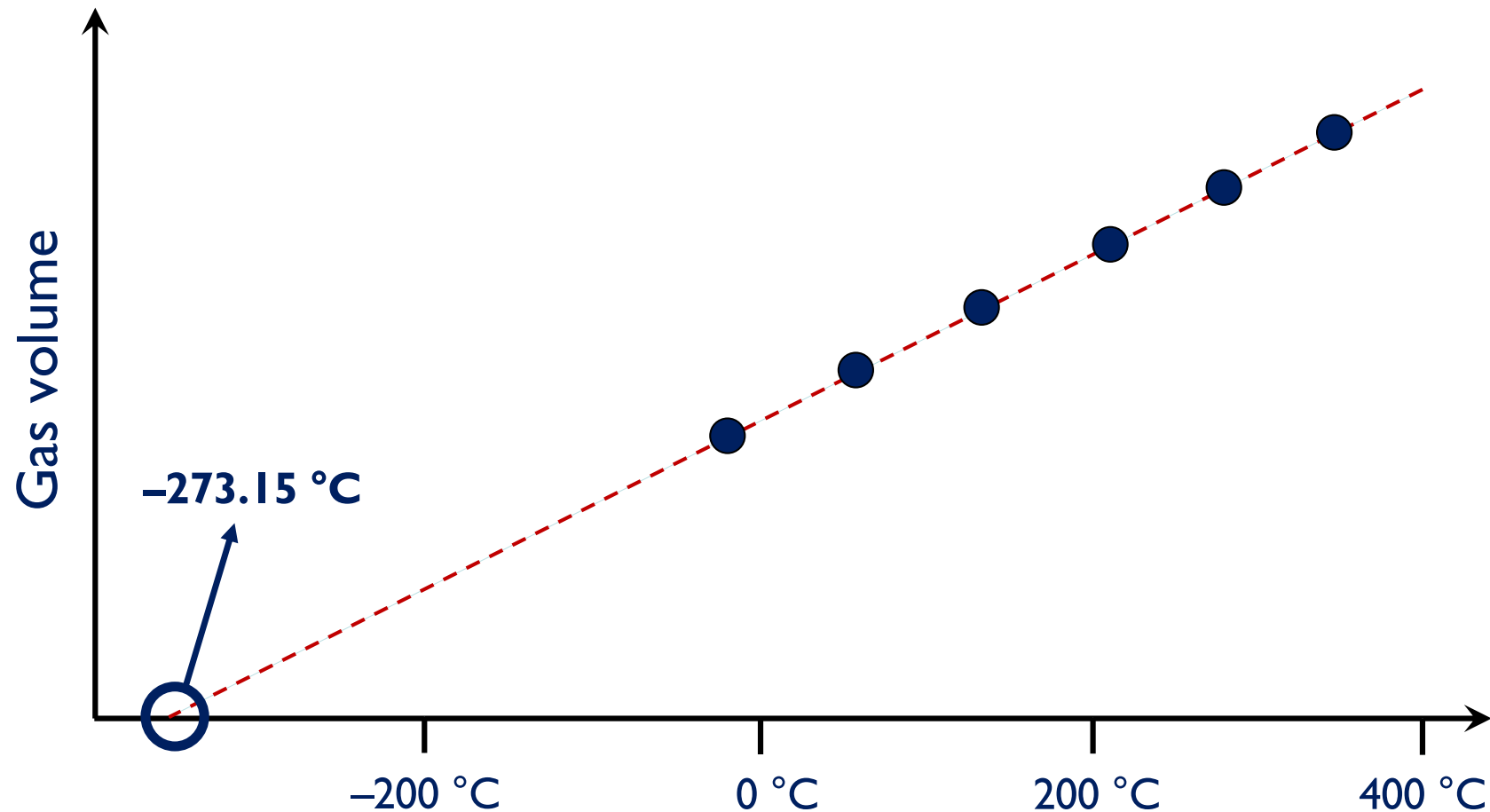
p – pressure; V – volume; n – number of moles; T – temperature

R – the gas constant: = 8.3145 J/(K mol)

= 0.083145 L bar/(K mol)

= 0.08206 L atm/(K mol)

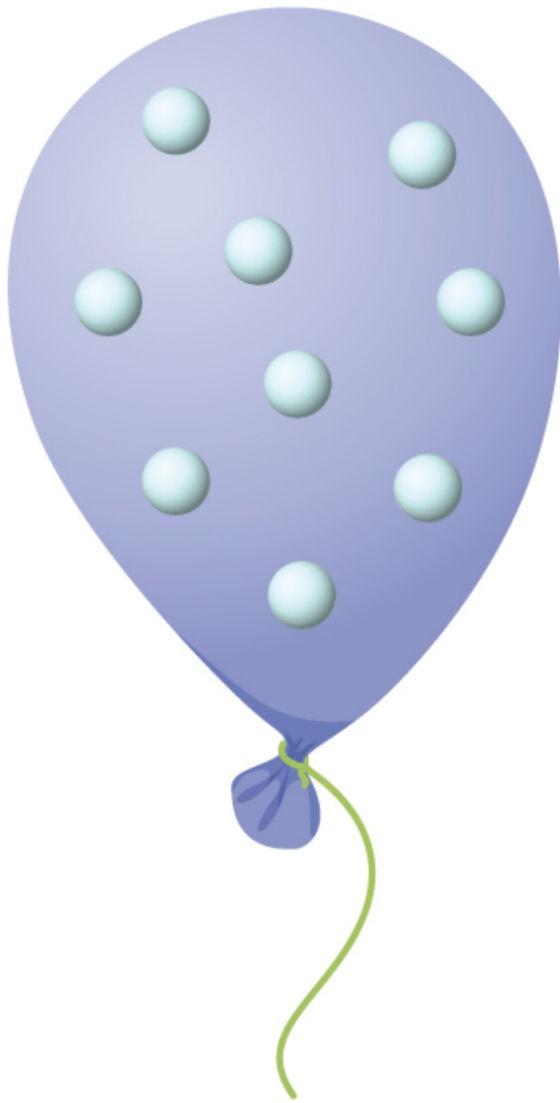
Absolute zero from Charles's law



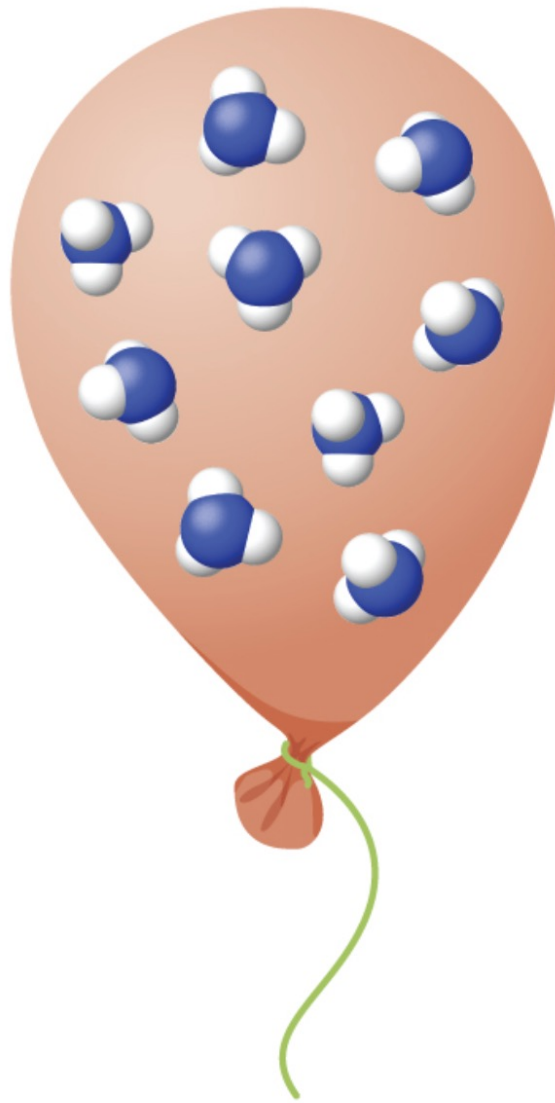
Extrapolation to $V = 0 \Rightarrow$ lowest attainable temperature

The absolute zero: $T = 0 \text{ K}$ (-273.15°C) (Lord Kelvin 1848)

Calculate the volume of 1.00 mol of ideal gas at room temperature (0 °C) and 1 atm of pressure. 🎉



He (4 g)



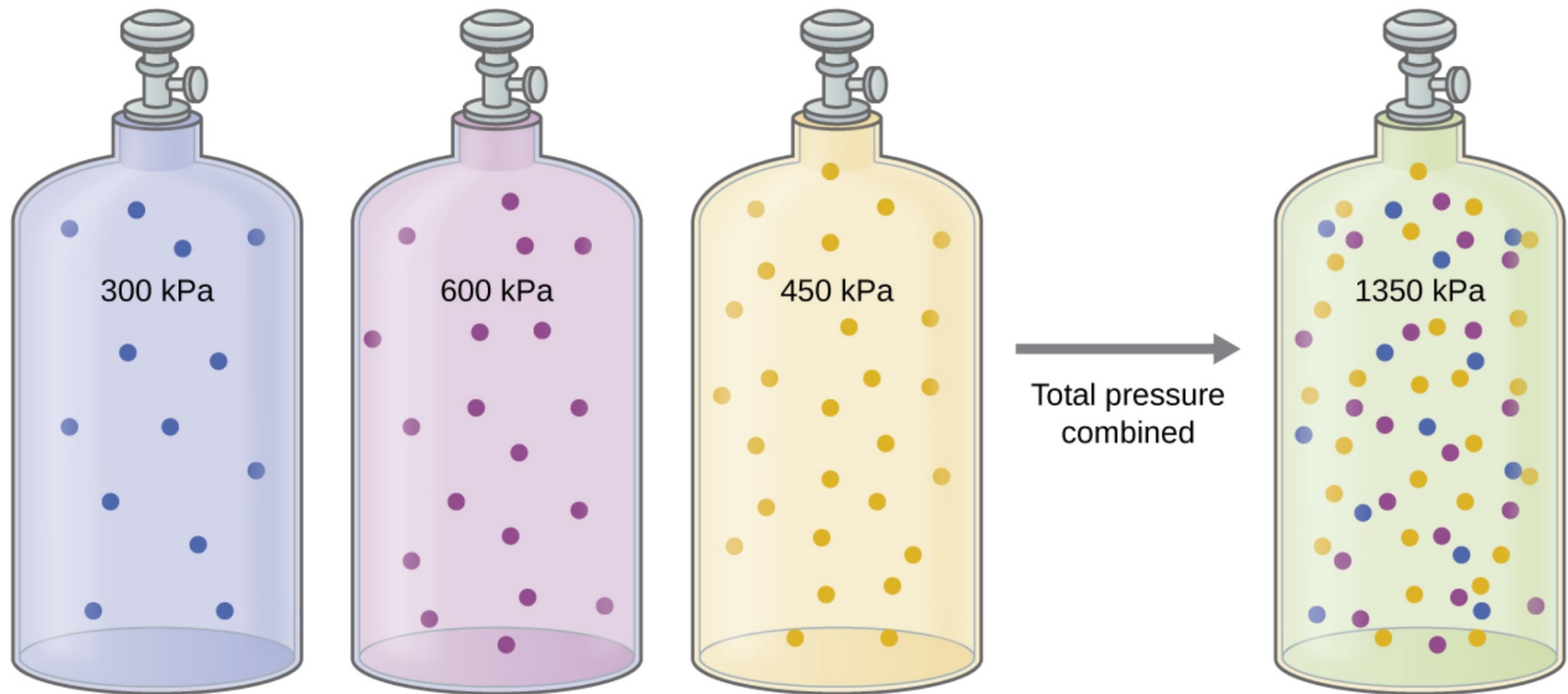
NH₃ (17 g)



O₂ (32 g)

Partial pressure

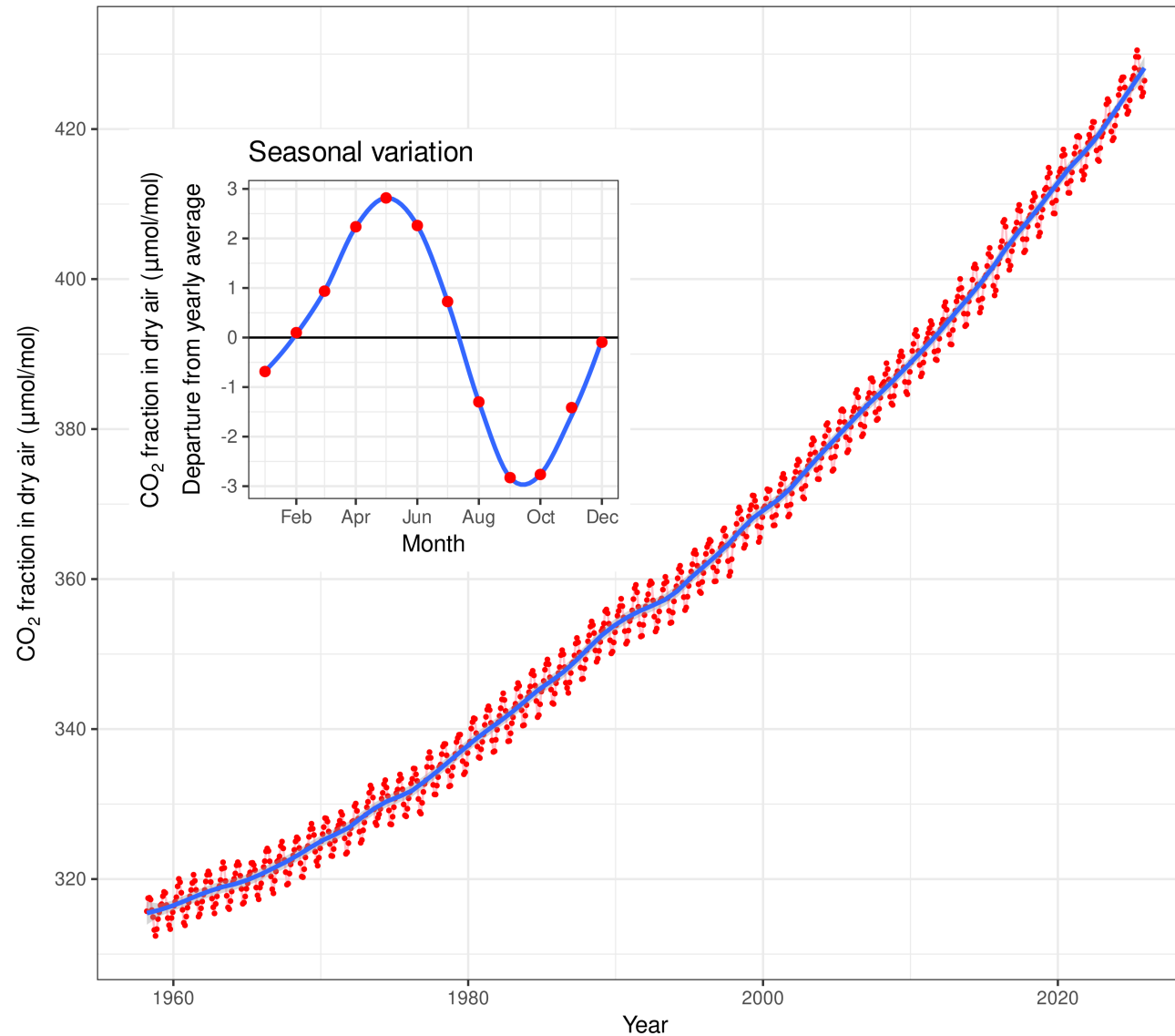
- Dalton's law (1802)



CO₂ in the atmosphere

Monthly mean CO₂ concentration

Mauna Loa 1958-2025



Data : Dr. Pieter Tans, NOAA/ESRL (<https://gml.noaa.gov/ccgg/trends/>) and Dr. Ralph Keeling, Scripps Institution of Oceanography (<https://scrippsco2.ucsd.edu/>). Accessed 2025-12-24
<https://w.wiki/4ZWn>

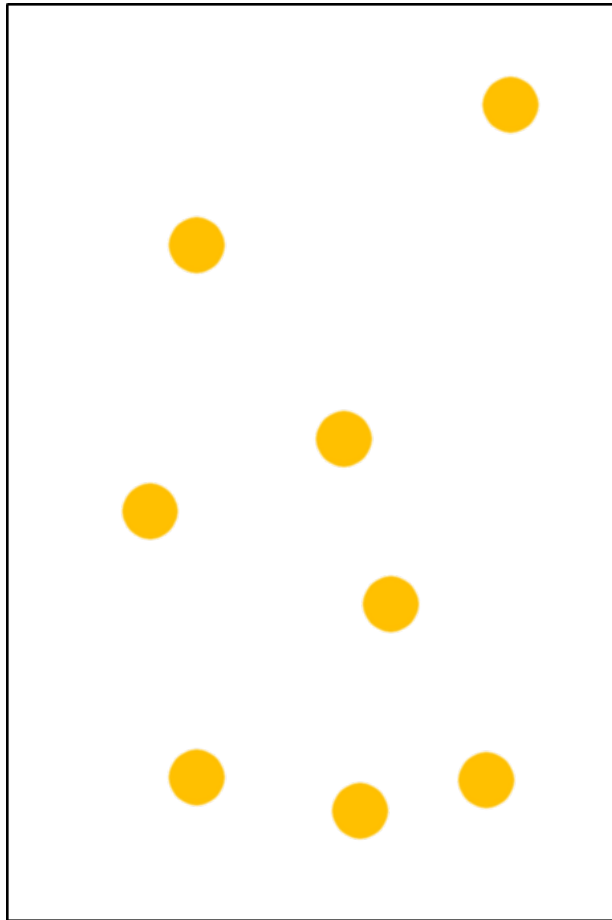
Ideal vs. non-ideal gas

ideal gas

p low and/or T high

small particle

negligible interaction

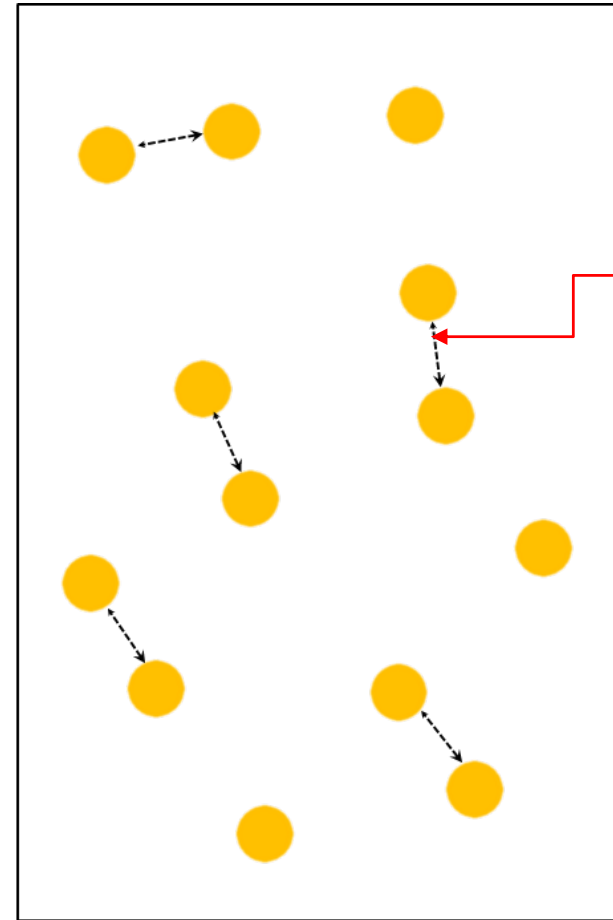


non-ideal gas

p high and/or T low

larger particle

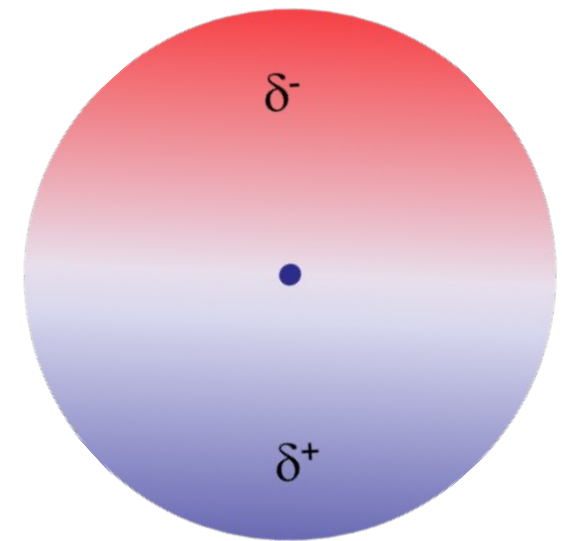
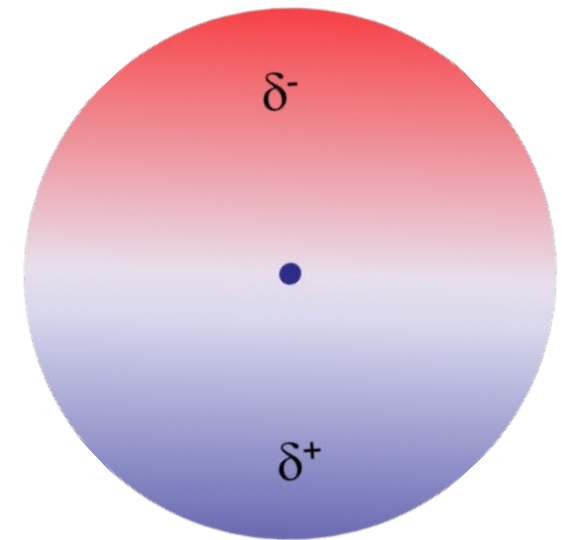
non-negligible interaction



intermolecular interaction

Energy of chemical bonds

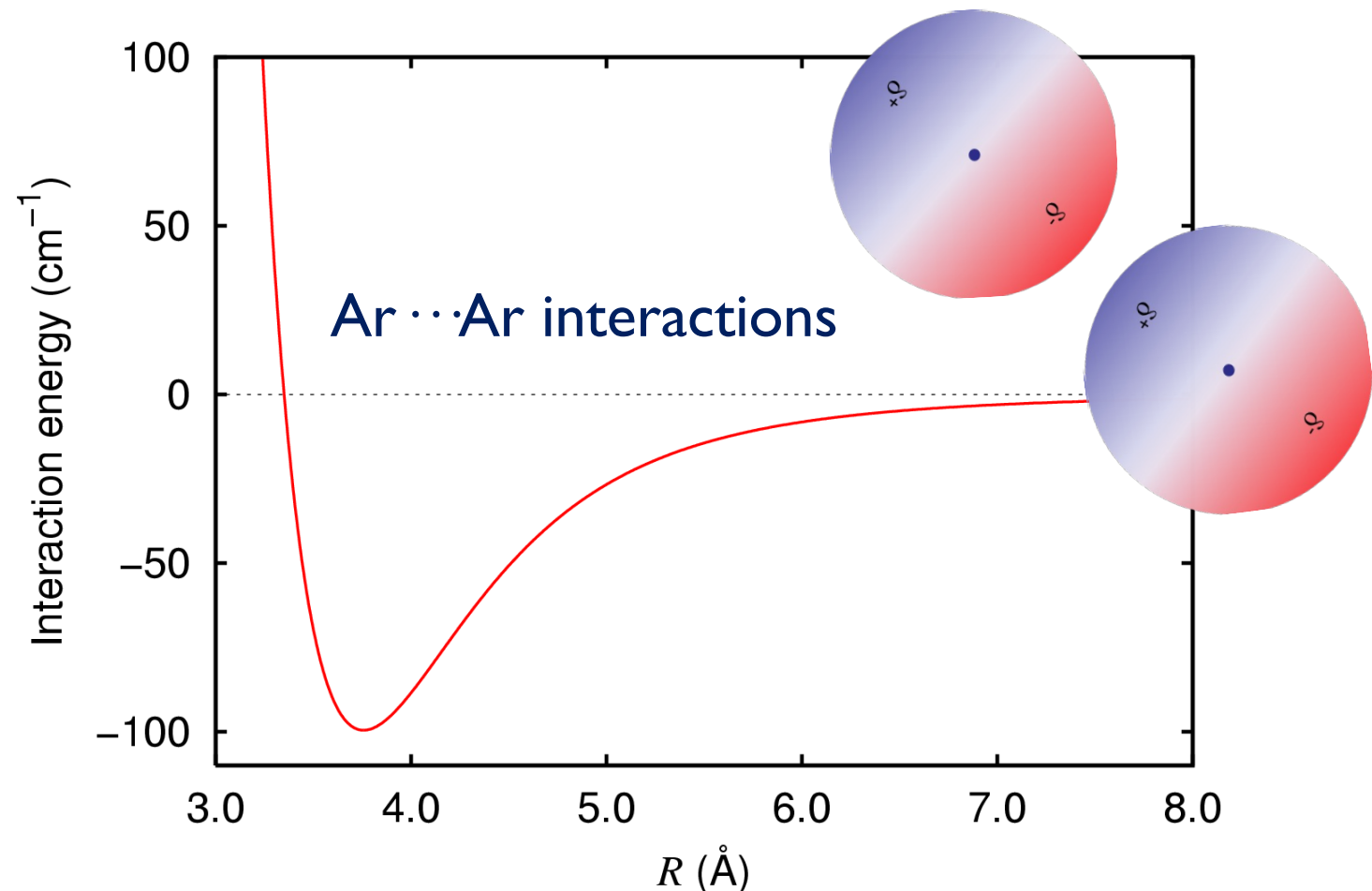
Bond type	Dissociation energy (kJ/mol)
Ionic lattice	1100–20000
Covalent bond	130–1100
Hydrogen bond	4–50
Dipole–dipole	2–8
London dispersion forces	<4 to 63



Real gases – van der Waals equation

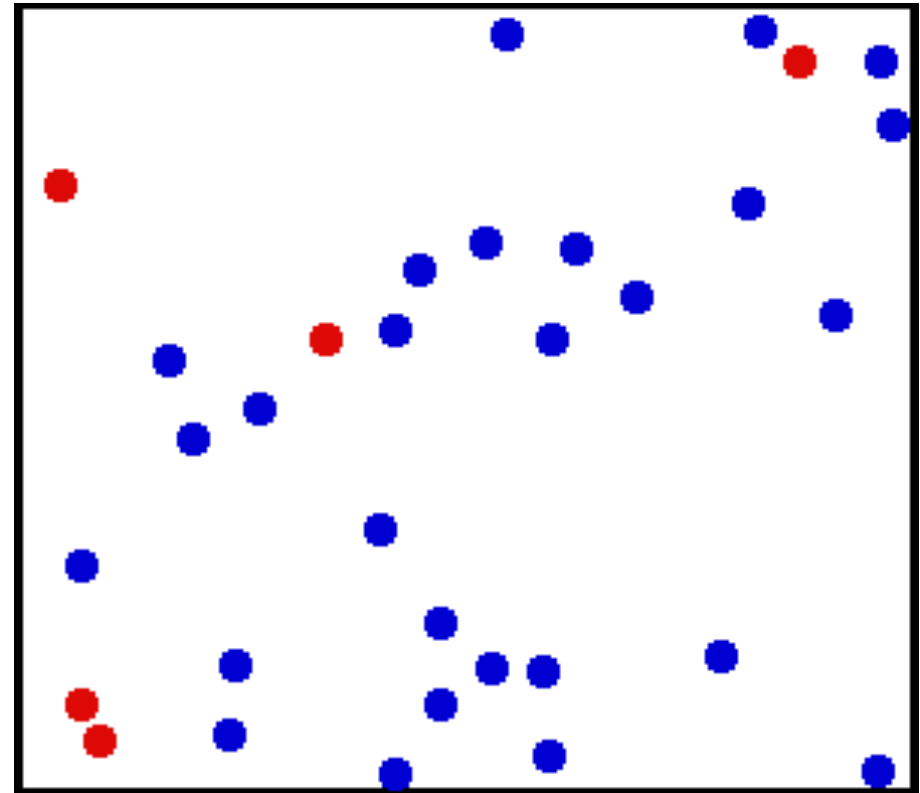
- $\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT$
- $\frac{an^2}{V^2}$ molecular interaction effect; nb finite molecular volume effect

$100 \text{ cm}^{-1} \approx 1 \text{ kJ/mol}$



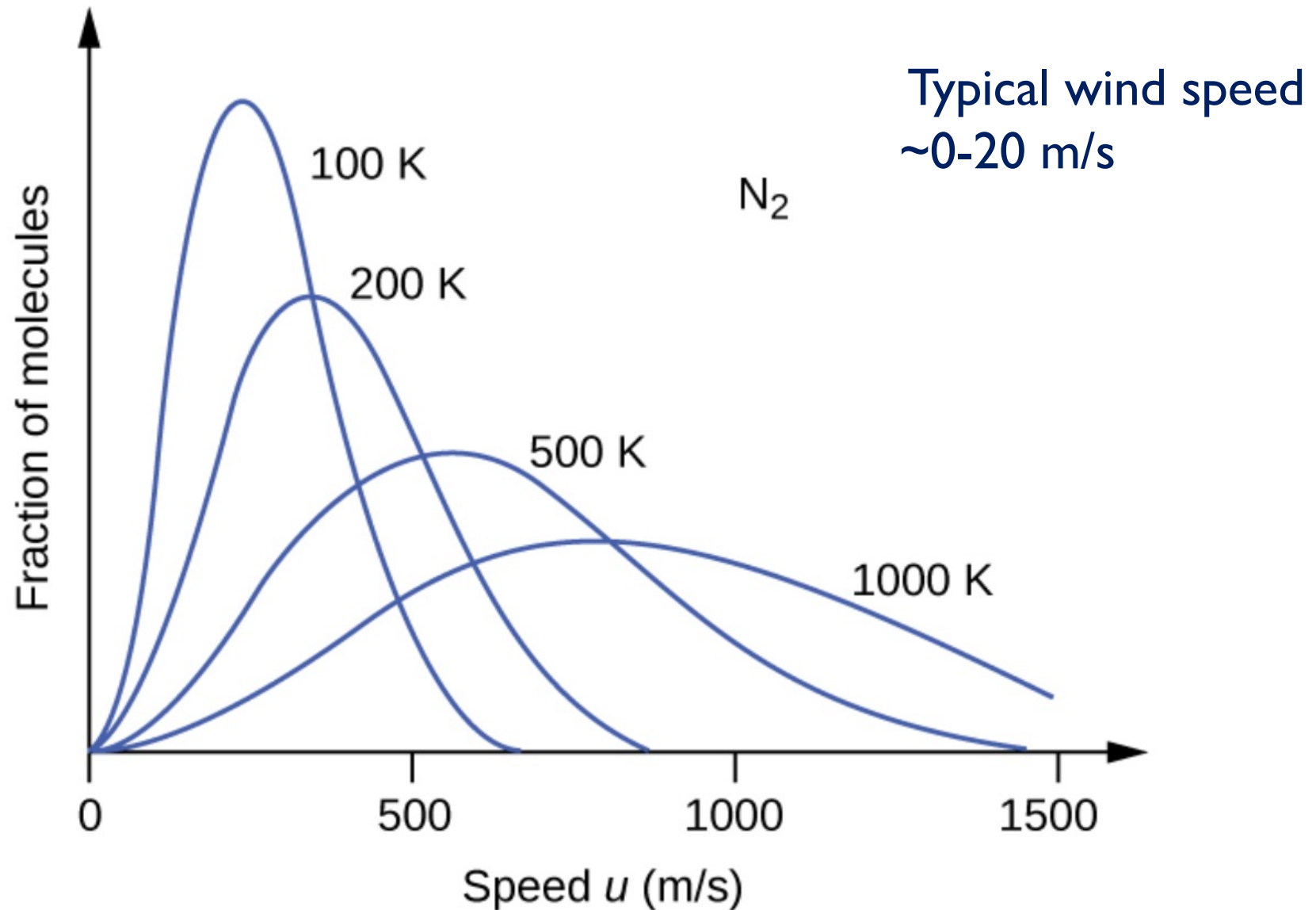
Kinetic gas theory of gases

- 1) Molecules are separated by distances much greater than their size. Can be considered as 'points'
- 2) Gas molecules are in constant motion in random directions. Collisions are perfectly elastic
- 3) There are no attractive or repulsive forces between gas molecules
- 4) The average kinetic energy of the molecules is proportional to the temperature (in K)



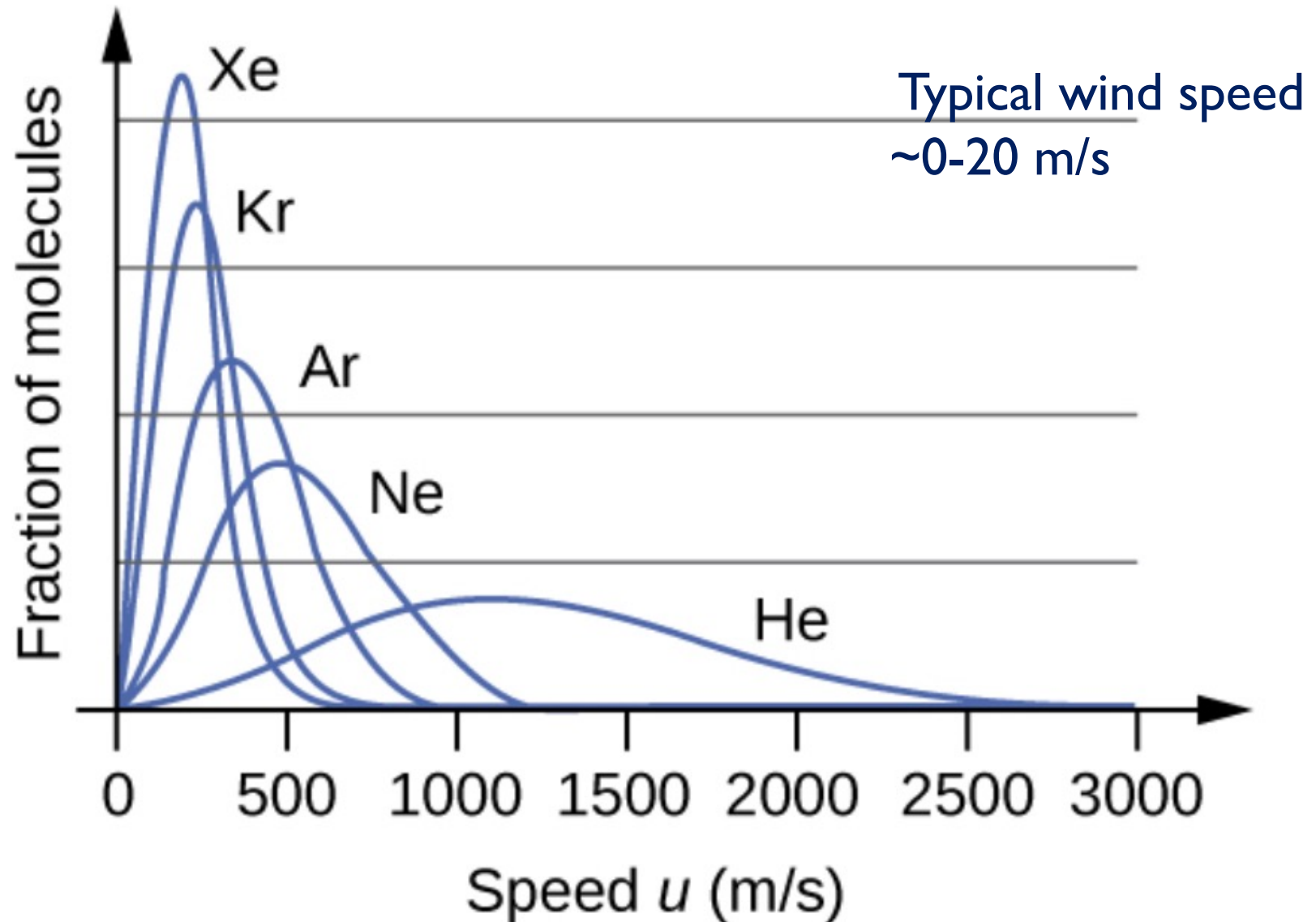
Kinetic gas theory

Maxwell-Boltzmann distribution

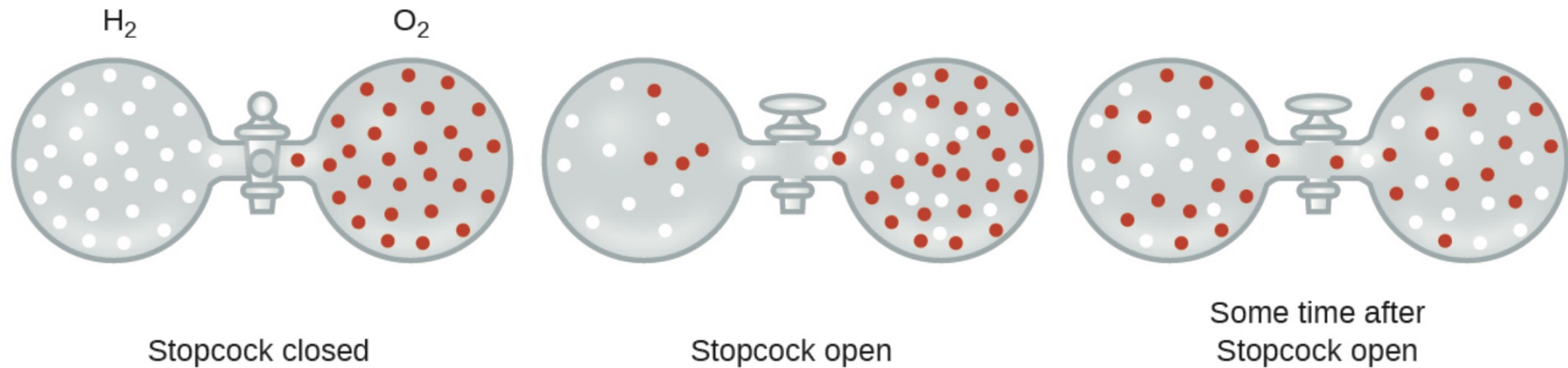


Kinetic gas theory

Maxwell-Boltzmann distribution

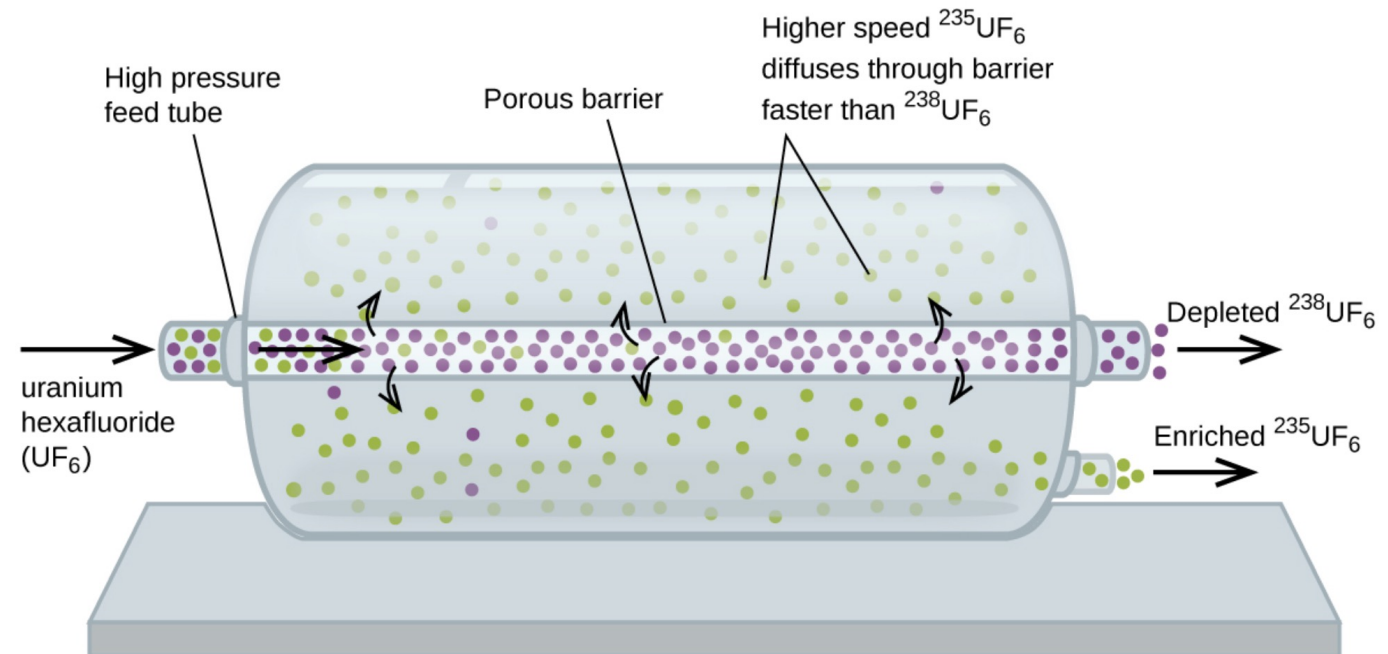


Gas diffusion



Rate of diffusion = amount of gas passing through an area/unit of time

From week 1:



Gas diffusion

Graham's law (r = rate)

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

DEMONSTRATION:

Calculate the ratio of diffusion rates of NH_3 and HCl